Heart Disease Prediction Using Machine Learning and Deep Learning Models

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Abstract— Early prediction of heart diseases which is also accurate is crucial for reducing worldwide morbidity and mortality rate. This study uses the Kaggle Heart Disease Prediction Dataset [1] to develop and evaluate a predictive pipeline for binary detection of heart disease. We implemented a comparative study using various machine learning classifiers (logistic regression, k-nearest neighbors, support vector machine, random forest, XGBoost) as well as fully connected deep neural networks, which we combined with preprocessing, feature selection and hyperparameter tuning. To ensure strong performance estimates, we trained our models using an 80/20 stratified split with cross-validation and evaluated using standard metrics (accuracy, precision, recall, F1-score, AUC-ROC). Among our used models, Random Forest achieved the highest accuracy of 0.9841, while our deep network achieved an accuracy of 0.9451. Our methodology is guided by previous studies that demonstrated robust performance of both ensemble methods as well as deep networks. We implemented similar preprocessing, feature selection and tuning strategies to assess their effectiveness on our selected dataset [2] - [10]. Our obtained results highlight the comparative advantages of classical machine learning and deep learning approaches on a medium sized dataset and provide a consistent baseline for future optimization. This study contributes to the continuous effort of improving automated prediction systems for cardiovascular diseases and underlines the importance of obtaining reliable results using careful preprocessing and model selection.

Index Terms— Heart disease prediction, Machine learning, Deep learning, Ensemble learning, Feature selection, Kaggle dataset, Hyperparameter tuning, Medical data analysis

I. INTRODUCTION

Cardiovascular diseases remain one of the leading causes of mortality worldwide, affecting millions of people each year. Early detection and accurate prediction of heart disease are therefore essential for improving treatment outcomes and reducing the global health burden. Traditional diagnostic methods are often time-consuming and can lack precision, which makes automated approaches an important research area. Recent advances in data science have enabled the use of machine learning (ML) and deep learning (DL) models for medical predictions. These methods can analyze large datasets, identify patterns, and improve diagnostic accuracy compared to conventional tools. In particular, ML algorithms such as Logistic Regression, Random Forest, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN), along with deep neural networks (DNNs), have been widely applied for heart disease prediction. This paper focuses on developing and evaluating a predictive pipeline using both machine learning and deep learning models on the Kaggle Heart Disease Prediction Dataset. By applying preprocessing, feature selection, hyperparameter tuning, and evaluation metrics, the study compares the effectiveness of classical ML approaches and DNNs in predicting heart disease. The aim is to establish a reliable and efficient model that contributes to clinical decision support systems.

II. LITERATURE REVIEW

Heart disease prediction has been a widely explored area in machine learning (ML) and deep learning (DL) research. Several studies have demonstrated the potential of computational approaches for early diagnosis while also highlighting challenges such as small datasets, limited generalizability, and model interpretability.

Singh and Kumar [2] compared multiple ML algorithms, including K-Nearest Neighbors (KNN), Decision Tree, Linear Regression, and Support Vector Machine (SVM), using the UCI heart disease dataset. Their results showed that KNN achieved the highest accuracy (87%), followed by SVM (83%). The main limitation of this study was the small dataset size of only 303 records, which restricted generalizability.

Sharma and Parmar [3] introduced a Deep Neural Network (DNN) optimized with Talos hyperparameter tuning. Their model achieved 90.78% accuracy, outperforming traditional ML approaches such as KNN (90.16%) and Logistic Regression (85.25%). However, due to reliance on a single small dataset, the model lacked external validation.

Bhatt et al. [4] employed a larger real-world dataset of 70,000 patient records. After preprocessing, classifiers such as Decision Tree, Random Forest, Multilayer Perceptron (MLP), and XGBoost were trained. The MLP achieved the best performance with 87.28% accuracy and an AUC of 0.95, showing that larger datasets significantly improve predictive robustness.

Al-Mahdi et al. [5] proposed an ensemble deep learning model with feature selection using a genetic algorithm and Tunicate Swarm Optimization. The hybrid approach achieved 97.5% accuracy, surpassing CNN-based baselines. While highly accurate, the model required extensive computational resources and was prone to overfitting.

Subramani et al. [6] developed a stacking-based fusion model combining ML and DL classifiers across five UCI heart datasets. Their approach achieved nearly 96% accuracy, outperforming individual models. Despite the improvement, the authors did not provide details on preprocessing or implementation constraints.

Chandrasekhar and Peddakrishna [7] emphasized hyperparameter optimization as a key factor in improving predictive accuracy. Bouqentar et al. [9] highlighted the role of feature engineering for early diagnosis, reporting strong performance using ML classifiers.

Finally, Bharti et al. [10] proposed a hybrid ML–DL model, combining classical classifiers with a fully connected deep network. On the UCI dataset, the ensemble achieved up to 94.2% accuracy. However, the study faced limitations from class imbalance, dataset bias, and lack of external validation.

Table I summarizes key contributions from recent works.

TABLE I

COMPARISON OF EXISTING HEART DISEASE PREDICTION STUDIES

Author(s)	**Dataset**	**Methods**	**Accuracy**	**Limitations**
Chandrasekar & Srinivasan	UCI (303 records)	KNN, SVM, Decision Tree, Linear Regression	KNN: 87%	Small dataset, limited generalization
Parmar et al.	UCI (303 records)	DNN with Talos	90.78%	No external validation, limited data
Aditya Khamparia &Dr. Mohammad Shabaz et al.	UCI (302) + IEEE Dataport (918)	Logistic Regression, KNN, SVM, RF, XGBoost , Deep learning	Deep Learning 94.2	Limited by dataset diversity
Saeed et al.	Kaggle (70,000 records)	Decision Tree, RF, MLP, XGBoost	MLP: 87.28%	Focused on one dataset only
Neeraj Varshney et al.	Multi-source combined (918 unique samples)	Stacking ML & DL models	96%(Approx.)	No preprocessing details, unclear constraints
Al-Mahdi et al.	UCI (76 attributes)	Ensemble DL + Genetic Algorithm + TSA	97.5%	High complexity, risk of overfitting

III. METHODOLOGY

A.Dataset Description:

The primary dataset that we used here is from Kaggle, a comprehensive dataset for Machine Learning-Based Heart Disease Prediction.[1] This dataset includes 14 features, which are commonly known to be factors in heart attack risk and also vital in predicting heart attack and stroke risks. It covers both demographic and medical characteristics. These 14 features contain a total of 1888 records collected from five publicly available cardiovascular disease datasets between 2018 to 2022. To ensure data reliability, missing data has been handled, too. The feature description includes pivotal features like age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels colored by fluoroscopy, and thalassemia. These features play a crucial role in the early prediction of cardiovascular disease.

B. Data preprocessing

All the preprocessing steps used were done in Python using Google Colab.

1. Data Loading

The CSV file of the dataset was loaded into a Pandas Dataframe. To inspect the structure and distribution of the dataset, basic exploratory operations were used (data.head(), data.info(), data.describe(), data.nunique()). After checking for missing values (data.isnull().sum()), it was found that the dataset did not contain any missing values.

2. Feature type identification:

In the code, after checking feature type,0 categorical variables and 14 numerical values were identified using data.select_dtypes().

3. Exploratory plotting:

For better feature understanding, various plots such as histograms, bar charts, and count plots were produced. For example, bar charts were produced for target, gender, chest pain, fasting blood sugar, resting electrocardiographic results,exercise-induced angina, slope, major vessels, thalassemia, and a histogram was produced for age.

4. Scaling:

As scaling is crucial for different models like Logistic regression, SVM, StandardScaler () was used to standardize all the differently scaled features. Then the scaled features were stored in X_scaled.

5. Split:

To split (X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42, stratify=y) was used and gives an 80%training and 20% test partition.

C. Learning phase (models, hyperparameter tuning, and training)

This study implements a mix of classical ML models and a feed-forward deep neural network. Among the used models, three were tuned using GridSearchCV (5-fold cross-validation).

Classical Machine Learning Models:

The classical models used were,

Logistic Regression (with max_iter=1000): A simple Linear Model that estimates the probability of a binary outcome using a logistic function.

Random Forest: An ensemble method where multiple decision trees are built and average their predictions to reduce overfitting and improve accuracy.

Support Vector Machine (with probability=True): A classification model which finds the optimal hyperplane that separates different classes of data points.

K Nearest Neighbor: A distance-based classification model which assigns labels based on the majority class of k closest training samples.

XGBoost (with use_label_encoder=False, eval_metric='logloss', random_state=42,verbosity=0): A framework of gradient boosting that sequentially builds models where each new model corrects the errors of the prior ones.

Hyperparameter Tuning was done for,

- Logistic Regression: (Hyperparameter grid: {'C':[0.01,0.1,1,10], 'penalty':['l1', 'l2'], 'solver':['liblinear']})
- Random Forest: (Hyperparameter grid: {'n_estimators':[100,200,500], 'max_depth':[None,5,10], 'min_samples_split':[2,5,10]})
- XGBoost: (Hyperparameter grid: {'n_estimators': [100, 200, 500], 'max_depth': [3, 5, 7], 'learning_rate': [0.01, 0.1, 0.2], 'subsample': [0.7, 0.8, 1.0]})

Hyperparameter Tuning is the process of systematically searching for the best set of parameters of the model that can maximize performance.

Deep Learning Models:

Deep learning models are composed of interconnected layers of neurons that can automatically learn complex feature patterns which makes them effective for tasks like binary classification and medical diagnosis.

The deep learning model was implemented using Tensorflow / Keras. The input layer was connected to a dense hidden layer of 64 neurons with ReLU activation. To reduce overfitting, a dropout layer was included with a rate of 0.3. Following this, another hidden layer of 32 neurons with ReLU activation was added. The network was concluded with a single output neuron with sigmoid activation to perform binary classification. For compiling the model, the chosen optimizer was 'adam', the loss function was 'binary_crossentropy', and the evaluation metric was 'accuracy'. Training was performed with a batch size of 32 out of 50 epochs using the training set, and validation performance was observed on the test set.

Random state was set to 42 in all experiments where applicable to ``improve reproducibility. For hypermeter selection, grid search used 5-fold cross-validation, and scoring was set to 'accuracy'.

D. Evaluation metrics and analysis

All the models were assessed on the test set. The primary metrics were Accuracy, F1-score, and ROC-AUC. To further evaluate classification performance, confusion matrices were visualized using confusion_matrix and ConfusionMatrixDisplay. The ROC curves were plotted for each model, and their AUC values were reported. To calculate the ROC points for the deep model, predicted probabilities were used (ann.predict(X_test).ravel()). Additionally, after identifying the optimal hyperparameters for LR, RF, and XGBoost, the best estimators were assessed on the test set, and their accuracy scores were noted.

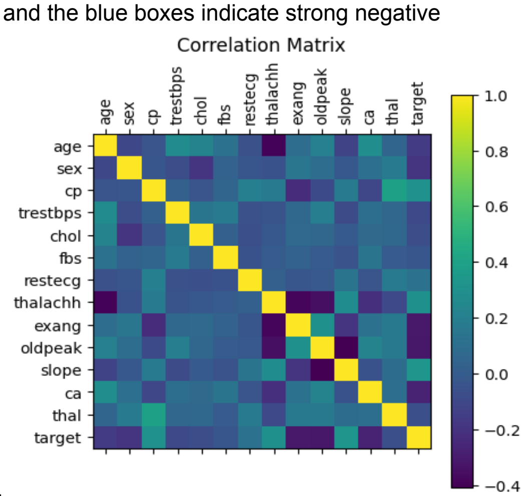
E. Experimental environment and computational resources

Here for an experimental environment and computational resources, an ANN (artificial neural network) was trained for 50 epochs with batch size=50 and verbose=1, and calculated epoch, loss,val_accuracy, and val_loss.

IV. RESULTS AND DISCUSSION

A.Feature Importance:

For a better understanding of the feature relation and patterns, we evaluated a correlation matrix and a corresponding heatmap. In the correlation matrix, the yellow boxes indicate strong positive correlations and the blue boxes indicate strong negative



correlations.

From the heatmap, it can be seen that, thalachh=0.3,cp=(0.3),exang=(-0.32),oldpeak=(-0.32),slope=(0.33) ,ca=(0.27) showed the strongest correlation with the target.On the contrary, all the other features showed the weakest correlation with the target.

B.Performance Metrics:

To evaluate the performance of the proposed model,we used five machine learning models

- ➔ KNN
- ➔ Random Forest
- ➔ SVM
- ➔ Logistic regression
- ➔ XGBoost

And some of the most frequently used performance metrics

- ➔ Accuracy
- ➔ F1 Score
- ➔ AUC

TABLE I
PERFORMANCE TABLE

Type	Model	Performance Metrics		
Classical Machine Learning Models		Accuracy	F1 (Weighted Avg)	AUC
	KNN	0.9233	0.92	0.96
	Random Forest	**0.9841** Tuned-0.976	0.98 Tuned- 0.975	1 Tuned-0.998
	SVM	0.9233	0.92	0.95
	Logistic Regression	0.7460 Tuned- 0.748	0.74 Tuned- 0.745	0.81 Tuned- 0.805
	XGBoost	0.9762 Tuned- 0.970	0.98 Tuned- 0.97	1 Tuned- 0.998
Deep Learnin g Model	ANN	**0.9451**	0.94	0.97

Accuracy is essentially the ratio of correctly predicted cases to the total number of cases, indicating the overall correctness of the model. On the other hand F1 score is the harmonic mean of precision and recall, which balances false positives and false negatives. Lastly, AUC evaluates how well a model can differentiate between all the thresholds.

Among the traditional machine learning models, the untuned Random Forest model had the highest weighted F1-score (0.98), overall accuracy (0.9841), and AUC of 1.0. Interestingly, hyperparameter tuning caused a slight decrease in all metrics (Accuracy = 0.976, F1 = 0.975, AUC = 0.998) rather than improving performance. With an accuracy of 0.9451, weighted F1 of 0.94, and AUC of 0.97, it is evident that even the untuned Random Forest outperformed the deep learning model (ANN) in this dataset. This suggests that for moderately sized structured datasets, ensemble-based classical methods may perform better than deep learning.

V. Conclusion

In this study, five classical models were benchmarked along with a feed-forward ANN on a Kaggle dataset with 1,888 records after doing standardized preprocessing, stratified splitting, and feature selection (for hyperparameter tuning). Across the measured metrics untuned Random Forest was the strongest performer (Accuracy 0.9841, F1-weighted 0.98, AUC ≈ 1.00), which exceeded the performance of the ANN (Accuracy 0.9451, F1-weighted 0.94, AUC 0.97). Although hyperparameter tuning was done, it did not improve the classical models. It is suggested from the results that on tabular, medium-sized data, ensemble methods can perform better compared to deep learning networks while also being simpler to interpret and train. Overall, our pipeline provides a reliable baseline that can predict the risk of heart diseases and a foundation for deployment-oriented refinements.

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